MAT9004 Assignment 2

Due: Wednesday, 30 April 2025, 23:55 PM

Please write up your solutions neatly! In particular, use notation and terminology correctly and include some necessary comments. Partial marks will be given for showing that you know some aspects of the answer, even if your solution is incomplete. The questions are NOT arranged in order of difficulty, so you should attempt every question.

Both handwritten and typed solutions are accepted. Please make sure you convert all your work into **one single pdf file** as other types might not display properly on moodle.

1. [8 points] Butane (C_4H_{10}) is commonly used as a fuel in lighters and camping stoves. When butane undergoes complete combustion in the presence of oxygen (O_2), it produces carbon dioxide (CO_2) and water (H_2O). That is, the unbalanced chemical equation for this reaction is

$$C_4H_{10} + O_2 \to CO_2 + H_2O.$$

To balance the reaction, the engineer needs to determine the correct stoichiometric coefficients (a, b, c and d) that satisfy the law of conservation of mass, which states that matter cannot be created or destroyed in a chemical reaction. This means that the number of atoms of each element must remain the same before and after the reaction.

   (a) [2 points] Using conservation laws for carbon (C), hydrogen (H), and oxygen (O), write a system of equations to balance the chemical equation. That is, the number of atoms of each element must remain the same before and after the reaction.

   (b) [1 point] Write the above system of equations in matrix form;

   (c) [4 points] Solve the above system using Gaussian Elimination and write your solution in a vector form.

   (d) [1 point] Hence, write the balanced equation.

2. [8 points] Consider the parabola $y = -x^2 + 4x - 3$.

   (a) [3 points] Find the tangents of the parabola at $x = 0$ and $x = 3$.

   (b) [5 points] Sketch the region bounded by this parabola and the tangents calculated in part (a) and find its area.

3. [7 points] During a dry season, water is gradually released from a reservoir to supply a nearby town. The release rate is $4e^{-0.03t}$ megalitres per day when water t days.

   (a) [2 points] Show that the rate is always positive and explain its physical significance in the context of water management.

   (b) [3 points] Find the total volume of water released, $V(t)$, as a function of time t.

   (c) [2 points] Describe the behaviour of V as $t \to \infty$, and determine what percentage of the total water release occurs in the first 60 days.

4. [13 points] A cybersecurity analyst is monitoring the spread of two types of malware, X and Y, within a corporate network. At each time step n, the number of infected machines follows a predictable pattern:

   - **If a machine is infected with Malware X:**
     - It infects **5** new machines with X.
     - It also infects **2** new machines with Y.
     - The original machine remains infected.
   - **If a machine is infected with Malware Y:**
     - It infects **3** new machines with X.
     - It infects **4** new machines with Y.

– The original machine remains infected.

Let x_n and $y_n$represent the number of machines infected with Malware X and Malware Y at time step n, respectively.

(a) [2 points] Find a 2×2 matrix A such that $\begin{bmatrix} x_{n+1} \\ y_{n+1} \end{bmatrix} = A \begin{bmatrix} x_n \\ y_n \end{bmatrix}$ for all integers $n \geq 1$.

To slow down the malware spread, cybersecurity measures are implemented. The new infection matrix (after intervention) is:

$$M = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}.$$

(b) [2 points] Compute M^{-1}, the inverse matrix of M. Suppose that 125 machines are infected with X and 125 with Y at time $n = 3$, use your result to determine the number of infected machines at time $n = 2$.

(c) [4 points] Find the eigenvalues and corresponding eigenvectors of M.

(d) [2 points] Find matrices P and D such that D is diagonal and $M = PDP^{-1}$, hence write the expression of M^n

(e) [3 points] Assuming the malware initially starts with one machine infected with X and zero with Y (i.e., $x_1 = 1, y_1 = 0$), estimate the number of machines infected with X and Y, and hence calculate the proportion of machines with type X virus. $x_n,\ y_n$, in the form $a \times b^n$ when $n$is very large.

5. [6 points] Consider the region representing the following relation:

$$\{(x; y) \in \mathbb{R}^2 : (x-6)^2 + (y-18)^2 \leq 361,\ (x+6)^2 + (y+18)^2 \leq 361,\ x - y \leq 0\}$$

Determine all relevant intersection points for the boundaries of the region, and where they intersect coordinate axes. Then provide a neat sketch the region. (Show full working of finding the intersections and keep the exact values.)

Total: 42 points